



Book I. Prop. VI. Theor.

In any triangle () if two angles ( and ) are equal, the sides ( and ) opposite to them are also equal.

For if the sides be not equal, let one of them  be greater than the other , and from it cut off  =  (pr. I.3), draw .

Then in  and ,
 =  (const.),
 =  (hyp.)
 and  common,
 $\therefore$ the triangles are equal (pr. I.4)
 a part equal to the whole, which is
 absurd;
 $\therefore$ neither of the sides  or
 is greater than the other,
 hence they are equal.

Q. E. D.